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CPSC 321 - Database Management
Professor Shawn Bower
Homework 5

1) Screenshots

a) List countries

```
Enter your choice (1-7): 1

United States of America (US), per capita gdp $21433, inflation rate 1.8%
Mexico (MX), per capita gdp $12600, inflation rate 4.2%
Canada (CA), per capita gdp $21433, inflation rate 2.1%
Thailand (TH), per capita gdp $5437, inflation rate 1.9%
Brazil (BR), per capita gdp $18000, inflation rate 4.2%

1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove Border
7. Exit

Enter your choice (1-7): █
```

b) Add country

i) Before Adding

```
dsosavida1_db=> SELECT * FROM country;
+-----+-----+-----+-----+
| country_code | country_name | gdp | inflation |
+-----+-----+-----+-----+
| US           | United States of America | 21433 | 1.8 |
| MX           | Mexico | 12600 | 4.2 |
| CA           | Canada | 21433 | 2.1 |
| TH           | Thailand | 5437 | 1.9 |
| BR           | Brazil | 18000 | 4.2 |
+-----+-----+-----+-----+
(5 rows)
```

ii) Adding

```
Enter your choice (1-7): 2
Country code.....: TR
Country name.....: Turkiye
Country per capita gdp (USD)...: 7000
Country inflation (pct).....: 4.9
Successfully Added Country
```

iii) After adding

```
dsosavida1_db=> SELECT * FROM country;
+-----+-----+-----+-----+
| country_code | country_name | gdp | inflation |
+-----+-----+-----+-----+
| US           | United States of America | 21433 | 1.8 |
| MX           | Mexico | 12600 | 4.2 |
| CA           | Canada | 21433 | 2.1 |
| TH           | Thailand | 5437 | 1.9 |
| BR           | Brazil | 18000 | 4.2 |
| TR           | Turkiye | 7000 | 4.9 |
+-----+-----+-----+-----+
(6 rows)
```

iv) Country already exists

```

Enter your choice (1-7): 2
Country code.....: MX
Country name.....: Mexico
Country per capita gdp (USD)..: 8000
Country inflation (pct).....: 4.2
Country already in DB!

```

c) Add border

i) Adding Border

```

Enter your choice (1-7): 3
Country code 1..: BR
Country code 2..: CA
Border length...: 3000
Successfully Added Border

```

ii) Before and after uploading BR and CA border

```

dsosavidal_db=> SELECT * FROM border;
country_code_1 | country_code_2 | border_length
-----+-----+-----
US             | CA             | 8893
US             | MX             | 3145
(2 rows)

dsosavidal_db=> SELECT * FROM border;
country_code_1 | country_code_2 | border_length
-----+-----+-----
US             | CA             | 8893
US             | MX             | 3145
BR             | CA             | 3000
(3 rows)

```

iii) Border Exists

```

Enter your choice (1-7): 3
Country code 1..: US
Country code 2..: CA
Border length...: 8893
Border already exists!

```

d) Find countries based on gdp and inflation

i) All countries

country_code	country_name	gdp	inflation
US	United States of America	21433	1.8
MX	Mexico	12600	4.2
CA	Canada	21433	2.1
TH	Thailand	5437	1.9
BR	Brazil	18000	4.2
TR	Turkiye	7000	4.9

(6 rows)

ii) Finding based on high inflation

```
Enter your choice (1-7): 4
Minimum per capita gdp (USD)...: 0
Maximum per capita gdp (USD)...: 99999
Minimum inflation (pct).....: 4.0
Maximum inflation (pct).....: 5.0

Brazil BR, per capita gdp $18000, inflation rate 4.2%
Mexico MX, per capita gdp $12600, inflation rate 4.2%
Türkiye TR, per capita gdp $7000, inflation rate 4.9%
```

iii) Mid-range GDP

```
Minimum per capita gdp (USD)...: 10000
Maximum per capita gdp (USD)...: 20000
Minimum inflation (pct).....: 0.0
Maximum inflation (pct).....: 5.0

Brazil BR, per capita gdp $18000, inflation rate 4.2%
Mexico MX, per capita gdp $12600, inflation rate 4.2%
```

iv) Not found

```
Enter your choice (1-7):
4
Minimum per capita gdp (USD)...: 10000
Maximum per capita gdp (USD)...: 20000
Minimum inflation (pct).....: 1.0
Maximum inflation (pct).....: 2.0
No countries found matching those criteria.
```

e) Update country's gdp and inflation

i) Altering Mexico GDP and inflation

```
Enter your choice (1-7): 5
Country code.....: MX
Country per capita gdp (USD)...: 10000
Country inflation (pct).....: 2.4
Successfully Updated Country
```

ii) Before and after alerting Mexico GDP and inflation

```
dsosavidal_db=> SELECT * FROM country;
country_code | country_name | gdp | inflation
-----+-----+-----+-----
US           | United States of America | 21433 | 1.8
MX           | Mexico | 12600 | 4.2
CA           | Canada | 21433 | 2.1
TH           | Thailand | 5437 | 1.9
BR           | Brazil | 18000 | 4.2
TR           | Turkiye | 7000 | 4.9
(6 rows)

dsosavidal_db=> SELECT * FROM country;
country_code | country_name | gdp | inflation
-----+-----+-----+-----
US           | United States of America | 21433 | 1.8
CA           | Canada | 21433 | 2.1
TH           | Thailand | 5437 | 1.9
BR           | Brazil | 18000 | 4.2
TR           | Turkiye | 7000 | 4.9
MX           | Mexico | 10000 | 2.4
(6 rows)
```

iii) Doesn't exists

```
Enter your choice (1-7):
5
Country code.....: ML
Country per capita gdp (USD)...: 5000
Country inflation (pct).....: 5000
Country does not exist!
```

f) Remove border

i) Removing border

```
Enter your choice (1-7): 5
Country code.....: MX
Country per capita gdp (USD)...: 10000
Country inflation (pct).....: 2.4
Successfully Updated Country
```

ii) Before and after removing

```
dsosavidal_db=> SELECT * FROM border;
country_code_1 | country_code_2 | border_length
-----+-----+-----
US              | CA              | 8893
US              | MX              | 3145
BR              | CA              | 3000
(3 rows)

dsosavidal_db=> SELECT * FROM border;
country_code_1 | country_code_2 | border_length
-----+-----+-----
US              | CA              | 8893
US              | MX              | 3145
(2 rows)
```

iii) Not found

```
Enter your choice (1-7): 6
Country code 1.: MX
Country code 2.: TH
Border doesn't exist!
```

g) Exit

i) Simple exit

```
1. List countries
2. Add country
3. Add border
4. Find countries based on gdp and inflation
5. Update country's gdp and inflation
6. Remove Border
7. Exit

Enter your choice (1-7): 7
(base) Davids-MacBook-Pro-2:hw-5 dauidsosa$
```

- 2) No serious challenges came up, except determining if there was already a duplicate. I had to learn new syntax in order to finish the assignment. Other than that, there were no difficult challenges, since I've built full stack applications using databases in the past.